

Laboratory Manual

Mine Ventilation Practical-II

(MNC 303)



DEPARTMENT OF MINING ENGINEERING

IIT (ISM), DHANBAD

General instructions on safety and Do's and Don'ts

- Come to the practical classes on time wearing cotton clothes.
- Keep your mobile phones in switched off mode in the laboratory.
- Listen carefully to your instructor and don't hesitate to ask question in case of any doubts.
- Do wear protective shoes, gloves and goggles while conducting the experiments to ensure proper safety.
- Don't touch or operate any laboratory instruments unless asked to do so.
- Do practice fire and electrical safety. Ensure that all electrical instruments are turned off before plugging in the sockets.
- In case any instrument is not functioning properly, bring it to the knowledge of your instructor.
- Read the laboratory manual carefully in advance, carry out the experiments as per the procedures laid down in it and record the observations in the appropriate format.
- Don't touch the hot surfaces of the equipment.
- Maintain safety distance from the instrument producing heat or flame.
- Keep the workplace clean.
- Handle the equipment with care and leave them in good condition after the experiments.
- Don't eat, drink or chew gums during the course of laboratory experiment.
- Don't leave any of your belongings in the laboratory and check before you leave.
- In case of any impending danger, evacuate the place and inform the concerned.

LIST OF PRACTICALS

Sl. No.	Name of the practical	Contact hours	Learning Outcome
1.	Determination of susceptibility of coal to spontaneous combustion using Crossing Point Temperature (CPT) method	2	Students will know the constructional features of CPT apparatus and method of determination of susceptibility of coals to spontaneous combustion using it
2.	Determination of Inflammability Index of a given coal sample using Godbert–Greenwald apparatus	2	Students will know the constructional features of Godbert–Greenwald apparatus and method of determination of inflammability of coal dust cloud using it
3.	Determination of airborne respirable dust concentration using Gravimetric Dust Sampler and Konimeter	2	Students will learn the constructional features of Gravimetric Dust Sampler and Konimeter, and method of measurement of airborne respirable dust concentration in mine atmosphere using them
4.	Determination of airborne respirable dust concentration using Personal Dust Sampler	2	Students will learn the method of assessment of personal exposure level of airborne respirable dust concentration in mine atmosphere using Personal Dust Sampler
5.	Determination of airborne respirable dust concentration using Real-time Aerosol Monitor	2	Students will learn the constructional features and method of measurement of airborne respirable dust concentration in mine atmosphere using Real-time Aerosol Monitor
6.	Study and sketch of self-rescuers (Chemical oxygen type and other models)	2	Students will know the applications, constructional features and working of different types of self-rescuers used in mines
7.	Study and sketch of self-contained oxygen breathing apparatus	2	Students will know the applications, constructional features and working of self-contained oxygen breathing apparatus used in mines
8.	Study and sketch of reviving apparatus	2	Students will know the applications, constructional features and working of reviving apparatus used in mines for artificial respiration of victims
9.	Evaluation of performance of rescue equipment using Artificial Lung Machine or Artificial Breathing simulator	2	Students will know the constructional features, working of Artificial Lung Machine and procedure for evaluating the performance of closed-circuit breathing apparatus using it

EVALUATION PATTERN

Sl. No.	Head	Distribution of marks
1	Quiz 1 & Quiz 2	40
2	Practical record	30
4	Viva-voce	30

Experiment-1

Aim: Determination of susceptibility of coal to spontaneous combustion using Crossing Point Temperature method.

Apparatus Required:-

- Crossing point temperature apparatus
- Thermometer
- Source of O₂ supply

Theory:

Spontaneous Combustion: Spontaneous combustion or spontaneous ignition is a type of combustion which occurs by the process of self-heating (increase in temperature due to exothermic internal reactions) of coal or other carbonaceous material, followed by thermal runaway (self-heating which rapidly accelerates to high temperatures) and finally, ignition.

The occurrence of spontaneous combustion in coal mines may be seen when the following three conditions are available:

1. Coal must be in a form which can oxidize readily **ate** ambient temperature.
2. Oxygen must be available to support the oxidation.
3. Favorable conditions for accumulation of heat must be available.

Crossing Point Temperature method: Crossing point temperature (CPT) is the temperature at which the coal temperature coincides with that of the furnace temperature or bath temperature in °C. CPT or critical oxidation temperature gives an idea about the proneness of coal to auto-oxidation or self-heating. This is a standard method followed in India for finding out the

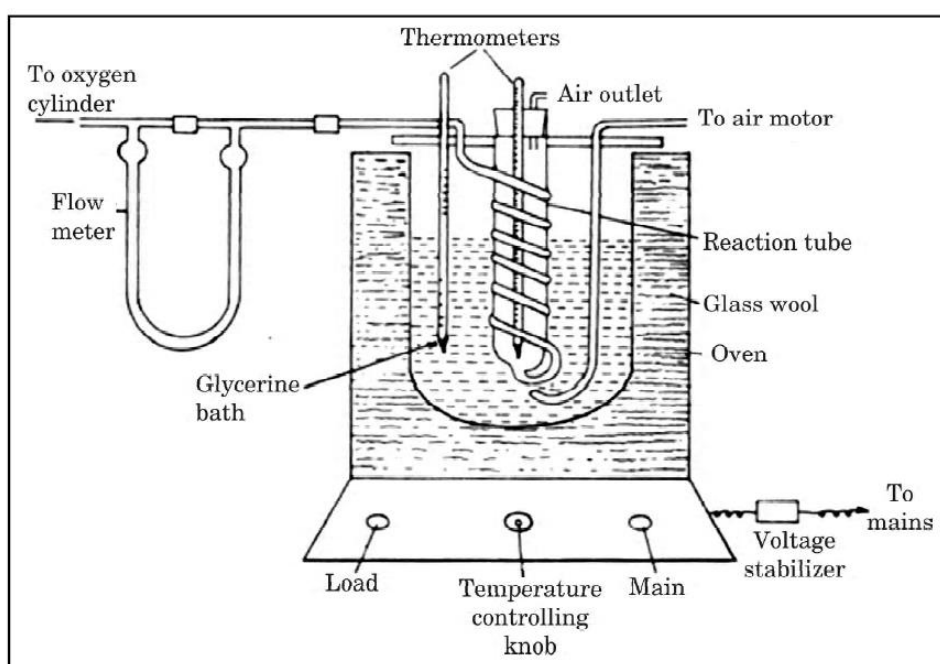
susceptibility of coal to spontaneous combustion. The higher the crossing point temperature value, the less will be susceptibility of coal to spontaneous heating.

Table 1: Risk rating criteria based on CPT values

CPT, °C	Spontaneous combustion susceptibility
120-140	Highly liable to self-heating
140-160	Moderately liable to self-heating
>160	Less liable to self-heating

This method envisages heating coal samples in an oxidizing atmosphere and in which oxygen/air is passed at a definite rate through a bed of coal powder immersed in a glycerin bath which is heated to raise the temperature at a constant rate. Initially the temperature of the coal bed shows lower value than the bath temperature. As the oxidation of coal is an exothermic reaction, it evolves heat and the temperature of coal bed increases gradually, oxidation rate is enhanced and a time comes when the temperature of coal bed equals that of bath temperature and then exceeds it. The point at which coal bed temperature equals the glycerin bath temperature is termed as crossing point temperature or critical oxidation temperature of the coal concerned. CPT depends on whether the bath is heated up at a constant temperature or at increasing temperature. Increasing bath temperature is considered to represent better simulation, as in actual cases of spontaneous heating, exothermic heat is accumulated by the coal and thus, the temperature of the surrounding increases gradually.

This is one of the oldest approaches for determining susceptibility towards spontaneous heating of coal. It is the temperature at which the coal temperature coincides with that of the furnace temperature or bath temperature in °C. In this method, the coal sample is heated in a furnace within a reaction tube at constant rising temperature with oxygen passing through it at a predetermined rate till the coal temperature crosses the furnace temperature.



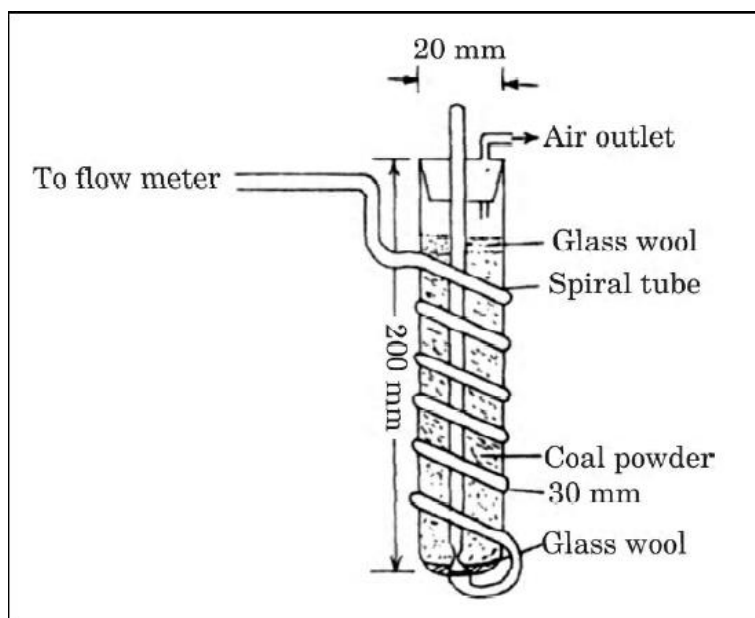


Figure: Photograph and schematic diagrams of the Crossing Point Temperature Apparatus

Procedure:

- 20 gm of coal sample of size -212 micron was placed in the reaction tube where glass wool is followed at the bottom most part of glycerin bath.
- The tube is then lightly tapped with the help of cork, a fixed number of times to achieve uniform packing density of the samples.
- Then the reaction tube is placed in the tubular furnace and a thermocouple is inserted at the centre of the sample for CPT experiment.
- After switching on, the furnace air is allowed to pass through the sample simultaneously with an average heating rate of 1°C per minute and air flow is at 80ml/ min. The temperature of the furnace and the coal sample are recorded at every 5 minute interval till the temperature of coal intersected over and gone beyond the furnace temperature as shown in the following figure.

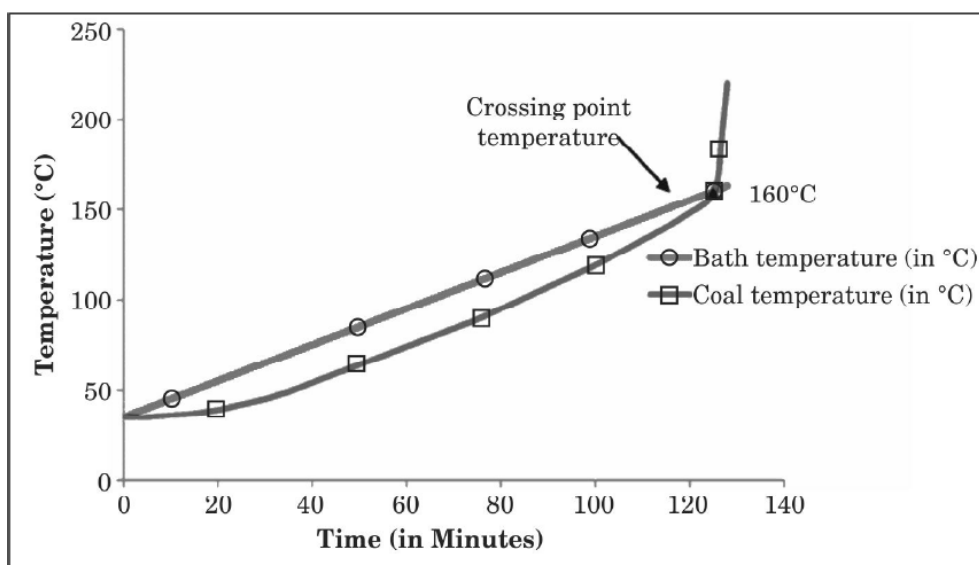


Figure: CPT for a coal sample

Observation table:

Time, min	Bath temperature, °C	Coal temperature, °C	Remarks
0			
5			
10			

Result: The temperatures of the glycerin bath and the coal sample are noted at equal time intervals and plotted on the graph paper.

Limitations: The major limitation of CPT experiment is that the rate of flow of O_2 , rate of heating, particle size of coal, *etc.* have to be properly maintained. Otherwise with slight change in any of the above conditions the CPT may change. This method is very time consuming and takes 3-4 hrs to complete experiment and failure of power supply even for a few minutes during experimentation especially when the temperature approaches CPT, gives erroneous results and the whole experiment is to be repeated.

Experiment-2

Aim:- Determination of Inflammability Index of a given coal sample by Godbert–Greenwald apparatus.

Apparatus Required:

- Godbert –Greenwald apparatus
- Oxygen Cylinder

Theory:

Flammability or explosibility index of coal

A unit weight of coal is dispersed in an environment of air or oxygen and ignited to cause explosion of coal dust. The amount of inert dust, which when mixed with the above coal dust, renders it non-flammable giving the measure of the explosibility of the coal seam. The explosibility factor or simply the explosibility of a coal-dust deposit (Ex) is given by

$$E_x = \frac{S}{100 - S}$$

Where S is the amount of total inert matter (per cent) that must be present in a unit mass of coal dust to prevent flame propagation through the mixture as determined by test. Depending on the explosibility index, the coal-dust explosions may be classified as follows:

Type of explosion	Explosibility index
Weak	< 0.1
Moderate	0.1-1.0
Strong	1.0-10
Severe	>10

Description of Godbert – Greenwald apparatus

The furnace comprises of a heated Silica tube capable of reaching temperatures of 1000 °C. Built-in control, trip and sample thermocouples are used to measure the internal temperature of the Silica tube. A sample powder injection module is close coupled to the furnace inlet and is mounted via the furnace sidewall assembly, for compactness. The bottom of the furnace is open to atmosphere allowing combustion products to vent into the fume cupboard. A stainless steel mirror is placed at a suitable angle and underneath the furnace to allow visual monitoring of the

sample as it enters the heated Silica tube. The furnace, stand and associated support assemblies are manufactured from stainless steel, whilst the sample powder injection module is manufactured from Brass.

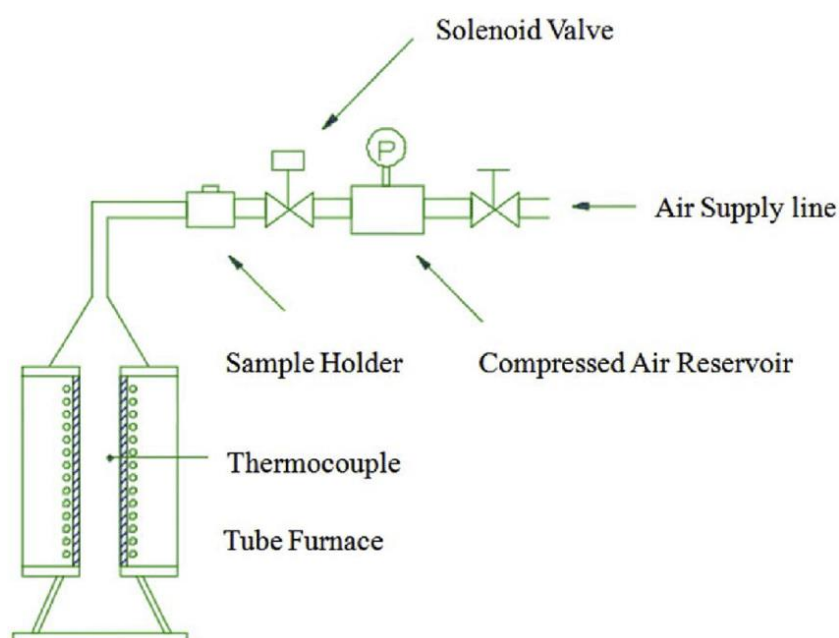


Figure: Photograph and schematic diagram of Godbert–Greenwald apparatus

Air Input Assembly

In order to inject the sample powder into the heated tube assembly a reservoir is pressurised via a regulated airline and then vented into the powder injection module. This pulse of air carries the sample powder into the heated Silica tube. The air input assembly comprises of an air reservoir complete with inlet / outlet valves and pressure gauge. This assembly can be assembled and customised for your installation as required.

Control / Monitoring Instrumentation

Control instrumentation for the Godbert-Greenwald furnace has been split into three modules, Temperature control / Air Reservoir Control, Heater Power Control and Remote Control Trigger.

Temperature Control: This module is responsible for setting and controlling the Silica Tube internal wall temperature. It also monitors over temperatures and automatically removes power from the heating elements of the Silica Tube if over temperatures are detected. The inside wall of the Silica Tube is also monitored, hence, the surface temperature of the tube is known. Eurotherm controllers and indicators are used for all control and display functions.

Air Reservoir Control: Manual control over the air inlet and outlet valves to charge and discharge the air reservoir can be carried out via this instrument. A timer function has also been employed to give greater control over the open time of the reservoir discharge valve, therefore allowing the valve to fully open or partially open to discharge the air reservoir.

Power Control Module: The power control module comprises of a mains transformer and solid-state relay, specifically for control of the Silica Tube heating elements. This module can be mounted locally to the furnace as required.

Remote Control Trigger Module: This module allows the operator to open the reservoir outlet valve whilst looking for an event inside the Silica Tube via the furnace mirror. A push button located on the remote trigger module is used to open the air reservoir outlet valve. Additional features located on the remote module allow the operator to manually charge the air reservoir and to adopt a semi-automatic mode where the air reservoir is charged with air after each venting.

INSTRUMENT OPERATION:-

- (i) Ensure that the control thermocouple is in thermal contact with the heater tube by observation only.
- (ii) Using the thermocouple extension leads connect the dual thermocouple in the furnace to the control input and sample input of the instrument. Also connect the single thermocouple to the trip input.
- (iii) Connect the transformer unit mains plug to HEATER OUTPUT protected mains outlet, next, connect the yellow 110v mains plug of furnace to socket.
- (iv) Switch the controller mains POWER SWITCH ON.

- (v) Apply power to the oven by pressing RESET push button after the required temperature trip level on 3216 trip temperature is being set.
- (vi) Once the furnace temperature has stabilized, switch on the air supply, set the pressure regulator to 5 psi and adjust VALVE OPEN TIME to 0.25s, next, check the operation of GG furnace remote control unit.
- (vii) The fume cup board interlock should be tested after installation and periodically thereafter.
- (viii) The equipment is now ready for minimum ignition temperature (MIT) dust cloud testing.

Observation Table:

% of stone dust	% of coal dust	Whether flame observed or not	Explosibility factor

Results and conclusion:-

The explosibility factor is calculated based on the percentage of stone dust in the coal dust-stone dust mixture at which no flame was observed.

Experiment-3

Aim: Determination of airborne respirable dust concentration using Gravimetric Dust Sampler (Model-MRE 113A) and Konimeter.

Apparatus required:-

- Gravimetric Dust sampler (Model-MRE 113A)
- Konimeter

Theory

MRE Gravimetric Dust Sampler:-The gravimetric dust sampler is a self-powered portable instrument for use in the gravimetric (by difference in weights) sampling of respirable airborne dust in mines. This is hung in a mine roadway at a height of $\frac{2}{3}$ rd of the full height of the roadway. The instrument operates for full shift and collects harmful dust particles on a filter.

From the difference in the weights of the filter, the harmful airborne dust per cubic meter of air is calculated.

A rechargeable 7.5 V, 90 mA battery drives the governed motor and a pump to draw 2.5 l/min of dust laden air through a horizontal elutriator size selector. The harmful dusts are collected on the filter. The amount of air sucked during the operation of the instrument is measured on a count, which registers the number of liters of air inspired by the instrument. The counter is resettable by the knurled accessible through the door of the instrument case. The pump is of diaphragm type and is fitted with two flap valves. A flowmeter has three horizontal lines engraved. The middle line represents the correct flow rate of 2.5 l/min and the one below represents an underflow of 2.5 l/min, whereas the above line shows an overflow of 2.6 l/min. The filters used are 5.5 cm circles of glass fiber type or membrane type. The membrane filter is much convenient to use.

$$\text{The amount of airborne respirable dust (mg/m}^3\text{)} = \frac{1000(W - w)}{N - n}$$

Where w is the initial weight of the filter (mg), W is the final weight of the filter (mg), n is the initial reading of the counter and N is the final reading of the counter. The amount of dust collected (mg) is $W-w$ and the air quantity sucked (litre) is $N-n$.

Description of the gravimetric dust sampler: The MRE 113A gravimetric dust sampler comprises basically a size selector, a filter holder and an electrically driven pump designed to pass 2.5 litres of air per minute. The coarse particles in the dust laden air settle down by gravity in the elutriator while the harmful fine particles in the respirable size range are collected on a filter of 55 mm in diameter. Glass fiber or membrane filters in a light aluminium holder are used for sampling. The amount of air in liters sucked during sampling is recorded on a counter. The counter is resettable to zero by the knurled knob through the door of the instrument case. The pump is of diaphragm type and is fitted with two flap valves and a flow smoothner at the outlet. A flowmeter is connected to the output side of the pump. The counter and the flowmeter are readable through small windows fitted in the side of the instrument. The window of the instrument has three horizontal lines engraved, the middle line representing the correct flow rate of 2.5 l/min.

The size selector is a precision made multi-channel horizontal elutriator mounted on the top of the instrument. A slotted restrictor plate is fitted at the mouth to minimize the effects of

air cross-flow, and the rear of each selector plate is lipped to prevent accidental shaking on to the filter of the dust collected in the size selector. The transfer hood between the filter holder and the size selector is held in position by two screws. Removal of this hood gives access to both the filter holder and the size selector for cleaning and blowing out accumulated dust.

Two interchangeable horizontal elutriators are available. The first with four channels provides a performance such that there is a 50% collection of particles with a falling speed equal to that of a 5 μ unit density sphere. The second with eight channels provides 50% collection of particles with a falling speed equal to that of a 3.5 μ unit density sphere.

The rear curved end of the instrument case is hinged and retained shut by a socket head screw, opening this end exposes the filter holder. The filter is fitted by unscrewing the knurled and winged ring, and inserting the filter into the gap between the upper and lower components until it seats in the recess. It is then clamped by screwing up the knurled ring. This causes the two halves of the holder to come together and grip the filter amount firmly the 'O' rings.

A small cylinder lock is fitted to the door of the case which when opened exposes the battery, counter and fuse holder. The battery is a DEAC 5/900D with a special internal limiting resistor added and it is held in its compartment by a lockable sliding spring clip.

The counter is coupled by gearing to the pump unit and registers the number of litres of air inspired by the instrument. It is resettable by the knurled knob accessible through the door of the instrument case. The pump is of the diaphragm type and is fitted with flap valves of aluminium coated polyester film. The pump output is varied by an adjustable crank and the unit is driven through super gears at half motor speed by a governed 6V 3-pole electric motor. A flowmeter is connected to the output side of the pump through a diaphragm type smoothing device. The counter and flowmeter are readable through small windows fitted in the side of the instrument case.

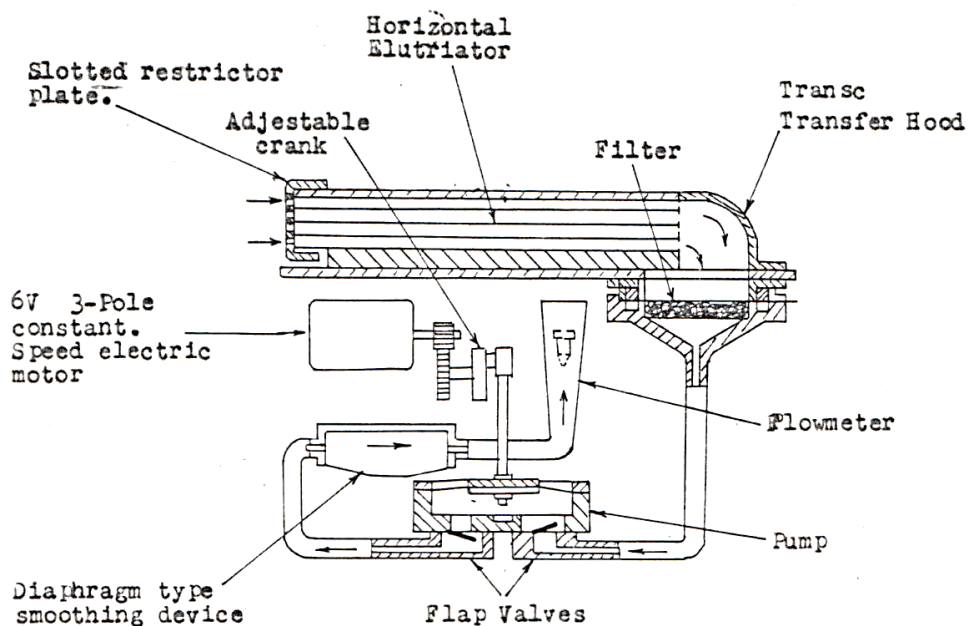


Fig. . Schematic diagram of M R E Gravimetric dust sampler.

Figure: Photograph and schematic diagram of Gravimetric Dust Sampler (Model: MRE 113A)

Procedure:

- A clean filter paper is selected and mounted in the filter holder. The assembly is weighed in an accurate balance having a sensitivity of 0.1 mg. It is inserted in the sampler and tightened in position. The battery is charged fully by the charger provided. A nearly discharged battery needs 14 hours charging. The charged battery is inserted in the instrument and the count is set to zero.
- The sampling instrument is then placed in the location where the breathable dust concentration is to be determined at a height approximately equal to the breathing level of the workers and switched on.

- Normally the duration for sampling covers a whole shift but the time can be less for high dust concentrations or more for low dust concentrations in order that sufficient amount of dust should be made on the rate of sampling which should be around 2.5 l/min. A lower rate of sampling indicates overloading of the filter paper with dust or discharged battery.
- After sampling is over, the volume of air sampled is noted and the dust-laden filter paper is removed from the instrument along with its folder and weighed. The difference in weight in mg divided by the volume of the air sampled in m³ gives the dust concentration in mg/m³.

Konimeter:

Description of Konimeter: The Konimeter consists basically of a spring-operated plunger pump, a dust deposition chamber, a glass plate, an air inlet jet and a built in microscope. Konimeter however can give a quick spot reading of dust concentration in p.p.c.c. (particles per cubic centimeter)

The components of the instrument are as follows:

- i. **Pump:** The spring operated pump sucks 5 cc of air sample. Gun metal plunger fitted with two rings moves to and fro in a cylinder. The plunger locks itself automatically while at the bottom of the cylinder.
- ii. **Dust Deposition Chamber:** The lower half of the chamber carries a glass plate on which the dust is deposited. It is possible to rotate this in 30 stages around the centre. On the upper portion, an air inlet jet is mounted through which the air sample is drawn into the dust chamber.
- iii. **Glass plate (50.8 mm × 4.34 mm):** The glass plate is a circular disc of heat resisting glass engraved with 30 sector markings. The plate rests at a fixed distance of 0.50 mm from the inlet valve and is made airtight by a rubber washer.
- iv. **Air Inlet Jet:** The air inlet jet consists of a carburetor type jet (0.5 mm dia) with parallel side passing through the upper portion of the dust chamber. This is protected by an outer rubber lined metal cap which is lifted only when sample is to be drawn in.
- v. **Microscope:** This is a simple biological type with a total magnification of ×216 (approx.). For rapid evaluation of the dust sample, a special graticule is fitted to the eye piece on which are engraved intersecting lines forming two 18° sectors together with two parallel lines 5 microns apart.

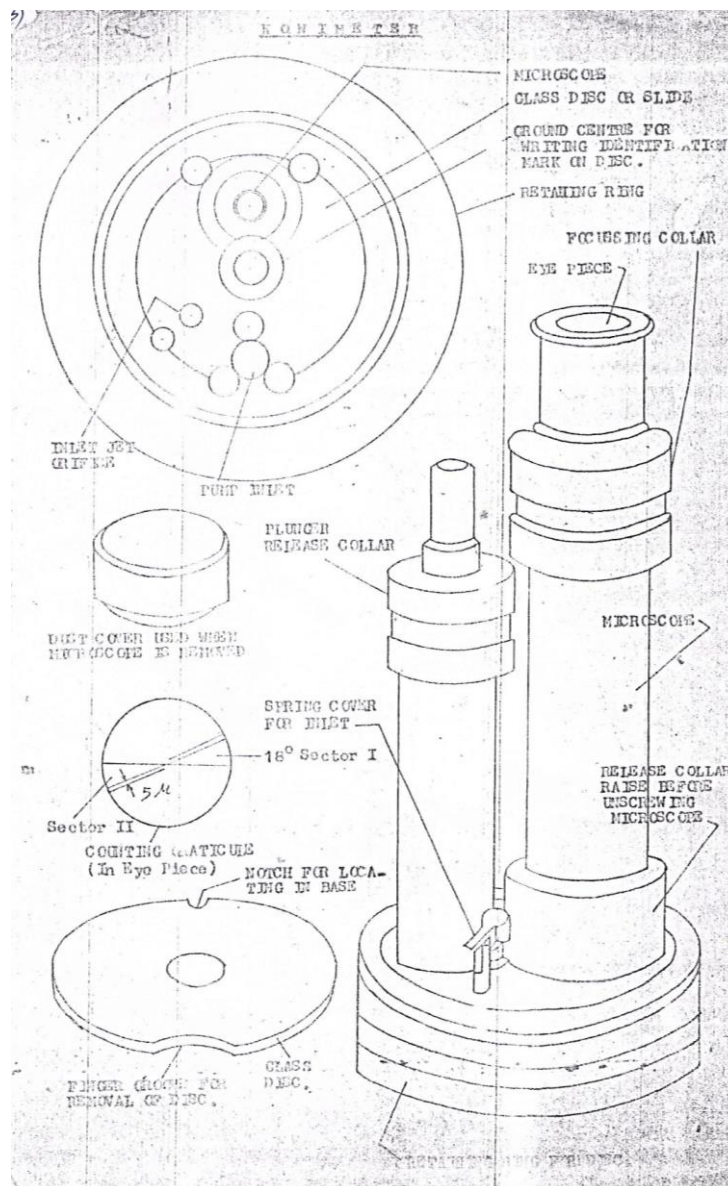


Figure: Photograph and schematic diagram of Konimeter

Procedure

1. Clean the glass plate with alcohol or some other organic solvents.
2. Treat the non-engraved side of the glass plate with a thin layer of petroleum jelly evenly. Replace the glass plate in the proper position.
3. Depress the plunger until it is locked at the end of down stroke.
4. Hold the Konimeter at the desired place and lift the air inlet cover.
5. Turn the knurled ring slightly to release the plate for automatic upstroke.
6. Collect the sample if desired, by rotating the plate.
7. The total dust count can be calculated by

- Total dust count in 5 cc of the air sample = $\frac{360 \times N}{36}$

- Total dust count in 1 cc of the air sample = $\frac{360 \times N}{36 \times 5} = 2N$

Limitations & Accuracy

- a) The performance or efficiency under varying conditions is unpredictable. Different types of Konimeter gives different results.
- b) It is difficult to evaluate a heavy deposit with high dust concentration as in blasting.
- c) The quantity of dust deposited depends on the nature and thickness of adhesive film.
- d) It has a tendency to break up aggregates due to high impact velocity.
- e) Only spot sampling is possible.

Observation tables:

For Gravimetric Dust Sampler

Sl. No	Initial weight of filter paper (w), mg	Final weight of filter paper (W), mg	Initial reading of flowmeter (n), litre	Final reading of flowmeter (N), litre	Dust concentration (mg/m ³)
1					
2					

For Konimeter

Samples	Dust count of less than 5µm size		Total
	N ₁	N ₂	N ₁ + N ₂
1			
2			

Results and conclusions:

- Using gravimetric dust sampler, the dust concentration in terms of mg/m³ can be calculated based on the initial and final weight of the filter paper, initial and final readings of the flow meter.
- Using Konimeter, the total dust count in terms of p.p.c.c. can be calculated by counting the number of dust particles deposited on the glass plate.

Experiment-4

Aim: Determination of airborne respirable dust concentration using Personal Dust Sampler.

Theory:

Principle: Personal Dust Sampler (CIP 10) is a compact, self-contained device for collecting dust samples in order to measure the personal exposures of workers to particular pollutants.

Construction: Personal Dust Sampler (CIP 10) is of plastic, elongated, monobloc design, comprising a cylindrical sampling head in the top part and in the bottom part, a flat case permitting it to be easily carried in the top pocket of a jacket. It has a height of 175 mm and the dimension of case is 100 × 70 × 25 mm. Its weight is 300 gm. Flat case contains the necessary parts for the appliance to function (i.e. motor, batteries, electronic regulating circuit, battery recharging plug and signal light indicating when the device is functioning). Figure shows the photographs and schematic diagrams of CIP 10. The sample inlet is symmetrical with respect to the vertical axis of the sampler. The sampling head, air mover and power pack are all combined into an integral package with no interconnecting tube. Sampled air is induced into the sampler by a sponge ring contained in a rotating plastic cup, which rotates at a speed of approx. 7000 rpm. Unusual pre-separation and sample collection methods are used. The pre-

separator consists of a small impactor, which removes the large particles entering the sampler, and a polyurethane foam sponge, which additionally removes a fraction of the non-respirable dust. Particles penetrating the foam sponge are collected by the denser rotating foam-sponge ring in the plastic cup. The penetration characteristics of the pre-separator are determined by the porosity and thickness of the sponge used. The unit has a sampling flow rate of 10 l/min which is controlled by the rotation rate of the sponge ring.

Polyurethane foam sponges are available with three pore sizes and in two thicknesses. A coarse sponge contains 45 pores per inch, a medium sponge contains 60 pores per inch and a fine sponge contains 100 pores per inch. Coarse and medium sponges are available in both 10 mm and 20 mm thicknesses while the fine grade sponge is supplied only in smaller thickness.

The sponge in the collection cup is a fine-grade foam material. Particulate collection is not absolute: therefore, a fraction of small particles passes through the sampler without being collected. The fraction is supposed to represent “respirable fraction” of the aerosol sampled.

Features of foam:

For top cup:

Grade	=	45 (Nominal pores per inch)
Thickness	=	5mm
External dia	=	23mm
Internal dia	=	7mm

Intermediate foam:

Grade	=	45
Thickness	=	24mm
External dia	=	20mm
Internal dia	=	8mm

Rotating cup foam:

Grade	=	60
Thickness	=	10mm
External dia	=	36mm
Internal dia	=	15mm

Weight of rotating cup = 3 g

Flow rate = 10 lit/min.

Rotating velocity = (6800 ± 200) rpm

Weight of total equipment = 300 g

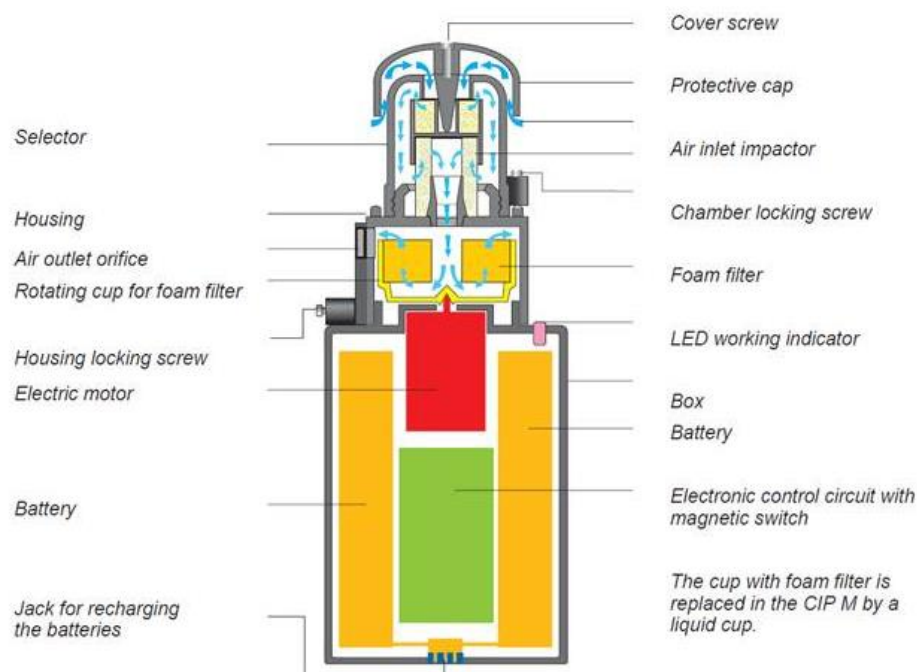


Figure: Photographs and schematic diagram of Personal Dust Sampler (Model: CIP 10)

Procedure:

The determination of the weight of the sampled respirable dust is achieved by weighing the rotating cup before and after sampling using a conventional laboratory balance. A balance accurate to 0.1 mg is generally sufficient. The difference between the two weighings gives the weight of dust collected.

Calculation

The average concentrations in mg/m^3 is calculated using the following formula:

$$C = \frac{P}{V}$$

Where P is the weight of sampled respirable dust (mg) and V is the total volume of filtered air (m^3).

Experiment-5

Aim: Determination of airborne respirable dust concentration using Real-time Aerosol Monitor (Model: RAM 1).

Apparatus Required: Real-time Aerosol Monitor (Model: RAM 1)

Theory:

The RAM-1 is a portable, self-contained aerosol dust monitor whose sensing principle is based on the detection of near forward scattered electromagnetic radiation in the near-infrared. The instrument uses a pulsed gas as semiconductor light emitting diode which generate a narrow-band emission centered on 940 nm. The scattered radiation is detected by means of silicon photo-voltaic-type diode with integral low noise preamplifier.

Signal processing is performed by employing a lock-in synchronous scheme which allows for continuous cancellation of detector and electronic circuitry drift and noise. The standard instrument is supplied with three selectable ranges (0-2, 20, 200 mg/m³) to allow for high resolution over a concentrated range of five decades. Further read out flexibility is obtained through four selectable time constant (0.5, 2.8, and 32 second); thus, the operator has complete control over both the range and speed of response.

The concentration data are continually displayed by a three and one half digit liquid crystal displayed. Included in the display are two diagnostics to alert the operator; a flashing "K" on the right hand side of the display indicates that the reference scatterer is inserted, and a flashing "VDC" indicates a low battery voltage. An analog voltage (0-10 VDC) output proportional to the concentration is also available for strip chart recording, data logging and /or telemetry.

The flow system employs a diaphragm pump to produce the desired sampling flow of 2 liters/minute. A secondary clean air stream of 0.2 liters/minute provide continuous flushing of filtered air over all critical optical surface. Also during a zero check, the entire inlet air flow is filtered to effect a self-cleaning of the optical sensing chamber. The flowmeter are provided to allow for continuous monitoring of the total and filtered flows. All filter cartridges are readily accessible and require no disassembly for replacement.

The instrument is designed to operate continuously for about 6 to 8 hours after its internal battery has been fully charged. A separate charger is provided with which the RAM-1 can operate indefinitely from the AC line.

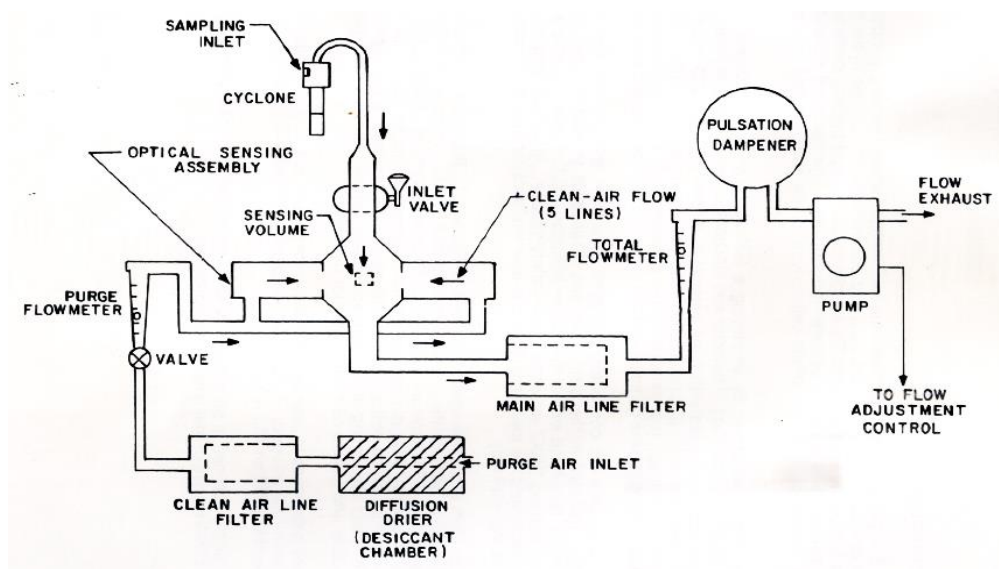


Figure: Photographs and schematic diagram of Real-time Aerosol Monitor (Model: RAM 1)

Measurement of airborne respirable dust concentration: The real-time Aerosol monitor (Model: RAM 1) directly measures the respirable dust concentration on real-time basis spot reading in following ranges:

1. 0 to 2 mg/m³
2. 2 to 20 mg/m³
3. 20 to 200 mg/m³.

Observation Table:

Sl. No.	Sample	Dust concentration, mg/m ³

Experiment-6

Aim: Study and sketch of self-rescuers (MSA IW-65 and other models)

Apparatus required: Self-rescuers (MSA IW-65 and other models)

Theory:

Self-Rescuer: The self-rescuer is a small emergency gas respirator designed to provide instant respiratory protection against **Co** gas which occurs in Mines **offer** an explosion or fire and thus enable the wearer to escape to a place **of safely**. It is compact and light in weight and is carried by the miner on his body. It should be used for entrance into an irrespirable atmosphere only when it is necessary to pass through it to get to an uncontaminated area self-rescuers do not offer protection in atmosphere containing less than 16% oxygen. They can be used for a period of 60min or more in concentration likely to be encountered after a fire or an explosion.

Filter Type Self-rescuer, MSA IW-65

Description: The IW65 Self-Rescuer Respirator provides emergency respiratory protection against carbon monoxide gas resulting from underground fires or explosions. The respiratory should always be carried so that it is ready for immediate use. The Self-Rescuer is a one-time device and must be discarded after use. Its weight is low, nearly 1 kg.

The respirator should be used immediately at the first sign of a fire or explosion-even if no smoke is visible. If the air is oxygen deficient (less than 19.5%), or there are other toxic gases and vapours present, the Self-Rescuer may not provide adequate protection. In such situations, a self-contained breathing apparatus must be used.

Principle of Operation: The self-Rescuer's filter section contains Hopcalite consisting of a mixture of manganese dioxide and copper oxide, an oxidation catalyst. This material changes toxic carbon monoxide into non-toxic carbon dioxide. The filter section also has an outer coarse-dust filter and an inner fine-dust filter to remove dust particles, and a drying agent to protect the Hopcalite from moisture. The filter materials are separated by screens and baffles. The drying agent is charcoal impregnated with a mixture of calcium bromide and lithium chloride in the proportion of 10:1 (approx. 53% CaBr_2 and 5% LiCl).

Inspired air is cooled by an integral heat exchanger before inhalation. Expired air passes back through the heat exchanger and out through the spring-loaded expiratory valve. Excess saliva is also expelled through the expiratory valve. The inhalation resistance at 95 l/min is approximately 6 mbar while the exhalation resistance is approximately 1.8 m bar.



Principle of operation

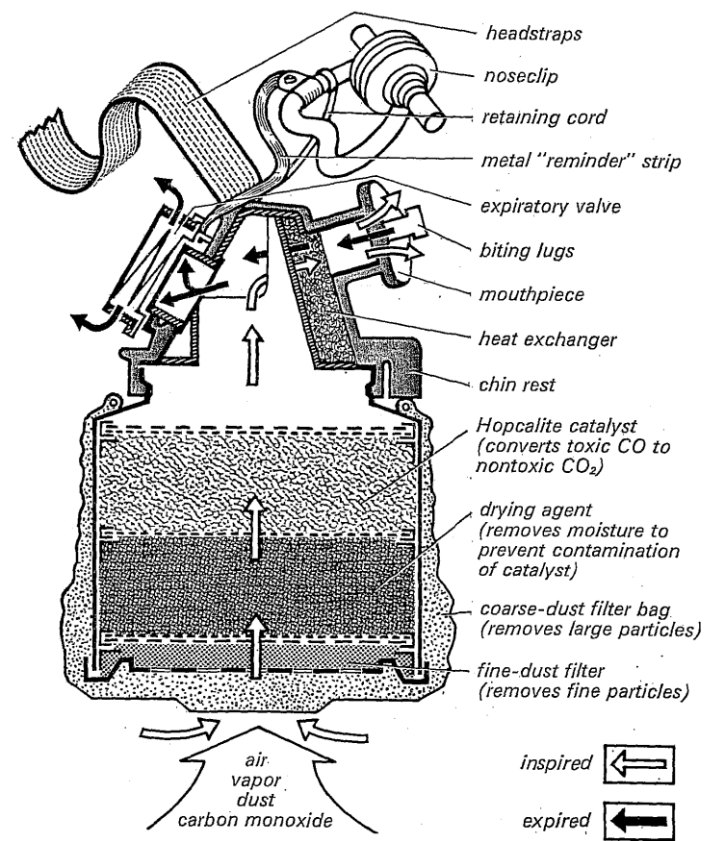


Figure: Photograph and schematic diagram of filter type self-rescuer (MSA IW65)

Drawbacks: The IW65 Self-Rescuer is an air-purifying device designed to protect the wearer from toxic carbon monoxide. It does not supply oxygen but functions to convert carbon monoxide to carbon dioxide with a resulting heat of reaction. Therefore, when the Self-Rescuer is worn in an atmosphere containing carbon monoxide, the air entering the wearer's mouth will be hot and dry. The temperature of the inhaled air is dependent upon the concentration of carbon monoxide – the higher the concentration, the higher the temperature. The very presence of heat indicates that carbon monoxide is present in the air being drawn into the Self-Rescuer, and the Self-Rescuer should never be discarded because of discomfort from hot, dry air. All air must be drawn through the Self-Rescuer.

The IW65 self-rescuer contains a heat exchanger to reduce the discomfort caused by high concentrations of carbon monoxide. For example, tests at 1.5% carbon monoxides showed that the heat exchanger will effectively reduce the temperature of inhaled air from approximately 149 °C to 65 °C. Though uncomfortable, one can tolerate even higher inhaled air temperature since the respiratory system itself is an effective heat exchanger.

Chemical type self-generating oxygen self-rescuer

Introduction:

Raksha Kavach is a single used self-contained closed circuit system oxygen type breathing apparatus. Raksha Kavach will assist a person to escape from irrespirable atmosphere containing smoke, toxic gases or as oxygen deficient atmosphere. As it is a closed circuit apparatus its operations does not depend on surrounding atmosphere.



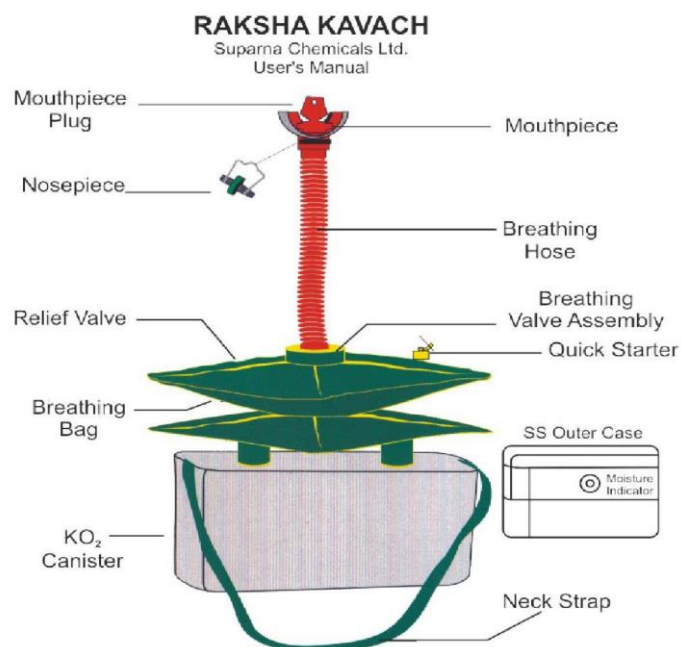


Figure: Photograph and schematic diagram of chemical oxygen type self-rescuer (Raksha Kavach)

(2) **Applications and Limitations:**

It is a life saving apparatus. It is designed for escape from any difficult and life threatening condition such as an environment of toxic gases, smoke & oxygen deficient atmosphere. It is used in emergency in:-

- (a) Underground mines, particularly in coal Mines.
- (b) Sub Marines
- (c) Many branches of industry

Raksha Kavach is designed for single use. It has got DGMS approval for 30min. duration.

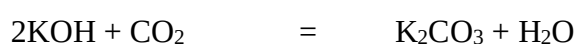
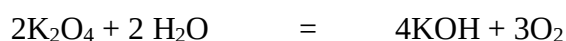
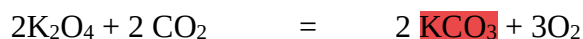
(3) **Design:**

Components:-

- (i) One stainless steel canister containing KO_2 .
- (ii) One 6 litres breathing bag with relief valve.
- (iii) One breathing valve assembly.
- (iv) One breathing hose with nose piece, mouth piece and mouth piece plug.
- (v) One neck strap.
- (vi) One heat protection &
- (vii) Suitable heat exchanger.

(4) **Principle of Operation:-**

The unit is activated with one or two breathes of the user. He inhales CO₂ and water vapour into the KO₂ belt of the apparatus through the breathing hose. KO₂ reacts with water vapour to produce oxygen and simultaneously remove CO₂ from the inhaled gas. The wear get oxygen free of CO₂. This chemical reaction is demand sensitive. The more frequently the user breaths the more water vapour and CO₂ gases inside the KO₂ bed & more oxygen is generated. At rest, breathing slows down and therefore oxygen generation is also less and KO₂ bed life is also more.



(5) **Operating Instructions:-**

- a) Remove unit from the belt.
- b) Lift the latch on the top.
- c) Remove bottom & top covers.
- d) Remove helmet.
- e) Put nose piece to close both nostrils.
- f) Remove mouth piece plug from mouth piece.
- g) Inhale ambient air through the mouth.
- h) Inhale into the mouth piece.
- i) Repeat the process 2 to 3 times to fill the breathing bag.
- j) Insert & hold mouth piece in bag.
- k) Breathe normally.
- l) Put on safety goggles.
- m) Adjust neck strips and hang breathing apparatus comfortably.
- n) Put on helmet.
- o) Move out.

(6) **Precautions:-**

- a) Raksha Kavach will be used by persons who are trained to use this equipment.
- b) The unit is a completely sealed unit and it is for single use only. Never open the latch of the unit out of curiosity. Its effectiveness may go down.
- c) Use only for escape in emergency.
- d) While using, do not remove mouth piece and nose dip unless you come to a safe place.

Experiment-7

Aim: Study and sketch of self-contained oxygen breathing apparatus (Model: BG-174A and BG-4).

Theory:

Dragger BG-174A

- It is a compressed O₂ type closed circuit apparatus with a constant and using governed dosage of O₂.
- Manufactured by Dragger AGCO, Lubbock, Germany.

Main Specifications:-

- Safe working period : 4 hr. hard work.
- Cylinder capacity (empty) : 2 ltr.
- Cylinder capacity with O₂ at 200kg / cm² : 400 lit.
- Breathing bag capacity : 6 lt. (approx)
- Weight of apparatus (fully charged) : 12.8 kg.
- O₂ flow rate : 1.5 lit/min.
- O₂ feed by lung governed value : As req. by weaser.



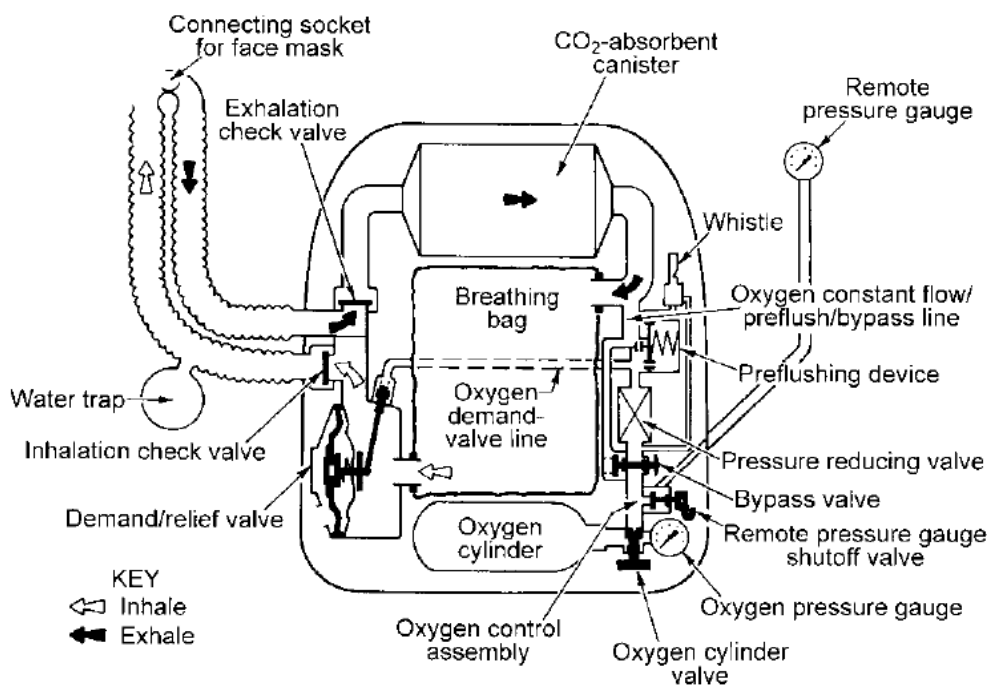


Figure: Photograph and schematic diagram of Drager BG 174A

Special features of the apparatus BG-174A:

- The O₂ regulator comprises a pressure reducer, an automatic pre flushing device, a bypass value, and a pressure gauge shut-off valve.
- The lung-demand valve is a diaphragm-operated stem valve in a valve box or housing and operated by the diaphragm responding to small negative pressures regulating from inhalation.
- A pre-flushing device fitted to automatically purge the breathing circuit of N₂ with a burst of about 7 lit. of O₂ when cylinder valve is opened.
- Apparatus is mounted on the back of the wearer.
- A dry ice cooler may be substituted for the saliva trap in the inhalation tube to cool the inhaled air temp to 15°C.
- Anhydrous NaOH (1.5Kg) is used as CO₂ absorbent.

Air circulating system:

- Exhaled air containing CO₂ and excess O₂ flows through the exhalation tube, exhalation valve into CO₂ absorption canister where it is freed from CO₂.
- The purified air flows into the breathing bag.
- On inhalation, it is drawn from breathing bag into long demand valve housing where it gets enriched with O₂ and then flows through the inhalation tube into the wearer's lungs.
- Excess pressure build up is prevented by the automatic relief valve.
- Additional O₂, if needed, is provided by the lung demand valve.

Draeger – BG-4

- It is a compressed – O₂, closed circuit apparatus with a disposable soda lime **CO₂** – absorbent canister.
- It has been NIOSH-certified four times (1995) as both a 3 & 4 hr. apparatus and a standard and positive pressure apparatus.
- The 4 hr., positive pressure version of the BG-4 was tested using the same O₂ cylinder as that used for the BG-174A, which was the same recommended fill pressure of 3, 135 psi at room temperature.
- The 3 hr. version comes with a fiberglass wrapped cylinder. The BG-4 has a constant flow of O₂ of at least **1-5** lit/min. ATPD plus a volume activated demand valve.
- The relief valve is also volume activated and purged exhaled air before it enters the **CO₂** absorbent canister.
- The face mask is the same as that used with the BG-174A.
- The wearer exhales into the face mask, through the exhalation check valve and breathing hose, through the **CO₂**-absorbent canister and into the spring loaded breathing bag.
- Upon inhalation, the user draws air from the breathing bag, through the heat absorbent canister, the inhalation hose and check valve, and back to the face mask.
- The heat-absorbent canister works by routing inhalation air around a chamber containing a large cylinder of ice inserted just before use, formed in a separate ice mold.



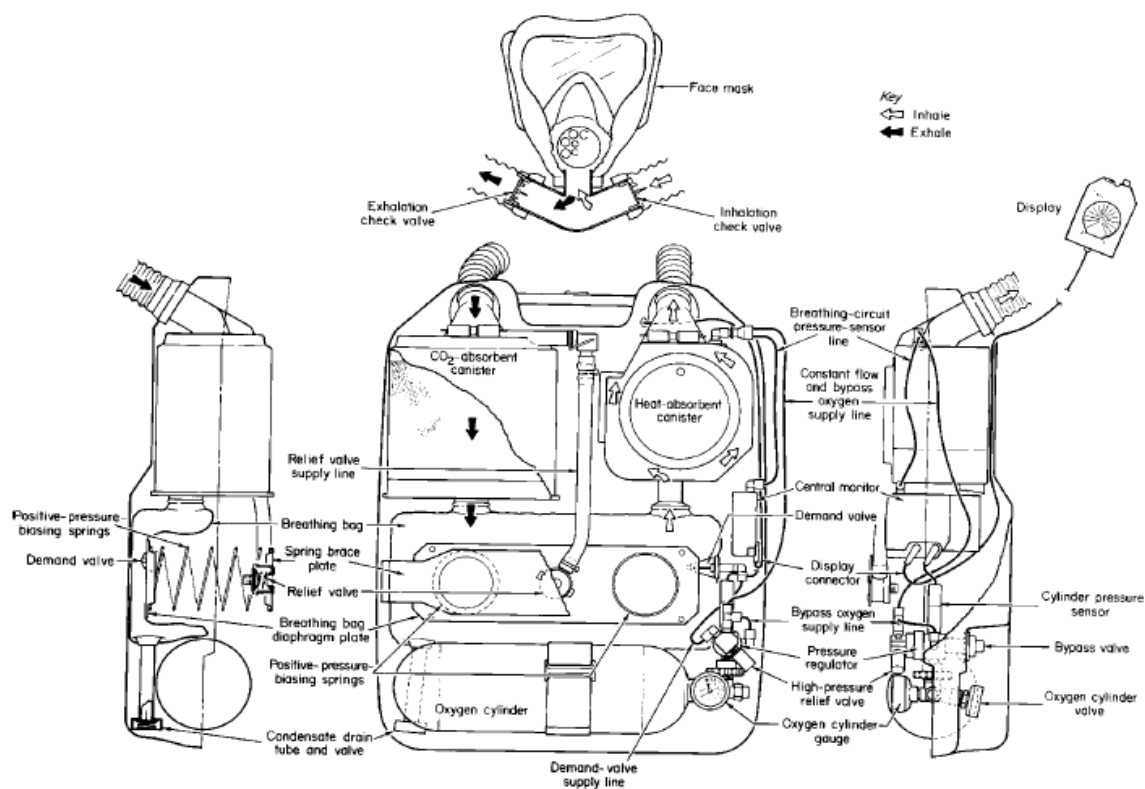


Figure: Photograph and schematic diagram of Drager BG 4

Experiment-8

Aim: Study and sketch of reviving apparatus (Model: Pulmotor PT-60).

Theory:

Introduction: The apparatus was first constructed by Herish draeger in a year 1906 & in course of time was considerably modified and improved upon. It provides for modified automatic cycling of breathing by blowing as oxygen-air mixture into the lungs of the victim and removing the Co2 charge exhausted air out of the lungs by suction with a simple adjustment, it can also be used as an inhalator.

Constructional features: It essentially consists of two 2.5 l capacity steel cylinder containing oxygen at a pressure of 200 kg/cm^2 , a pressure reducing valve a distribution connection or manifold with three shut-off valves and a safety valve, a resuscitation or change over or cycling valve box, a face piece, an aspiratory injector, a secretion bottle, an oxygen economizer breathing bag, an inhalation face piece and three sizes of oral airways. The reducing valve reduces the high pressure oxygen to a constant operating pressure of 5 kg/cm^2 of the three-shut

off valve, one is connected to two aspirating injector, another to the oxygen inhalation mask, while the third **id** connected to cycling valve box. The cycling valve box is the soul of the apparatus.



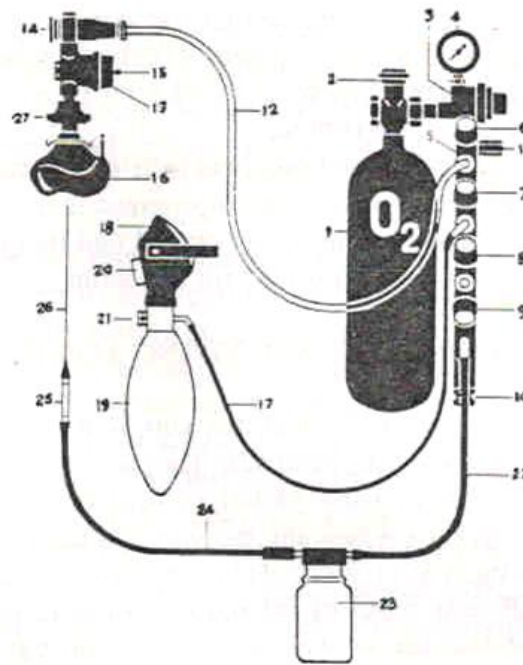


Fig. 7.28: Draeger Pulmotor 1 - oxygen cylinder; 2 - cylinder valve; 3 - reducing valve; 4 - pressure gauge; 5 - distributor manifold; 6 - pulmotor shut-off valve; 7 - inhalation shut-off valve; 8 - spare valve for connecting other equipment; 9 - suction valve; 10 - aspirator injector; 11 - safety valve; 12 - oxygen connecting hose; 13 - pulmotor switching unit; 14 - change-over valve for 50% or 100% oxygen; 15 - selector knob; 16 - facepiece; 17 - connecting inhalation tube; 18 - inhalation facepiece; 19 - economizer breathing bag; 20 - expiratory valve; 21 - air addition valve; 22 - connecting hose; 23 - secretion bottle; 24 - aspirator hose; 25 - secretion vision glass; 26 - aspirator catheter; 27 - respiratory valve (for cutting-off the negative phase), optional

Figure: Photograph and schematic diagram of Pulmotor PT60

Operating Process: During the inspiration phase, **oxygen** flows through an injector nozzle into the upper part of the valve box there by sucking atmospheric air through a spring loaded air regulating valve. The oxygen air mixture thus obtained flows through the floating needle valve into the lower part and there to the victim. As soon as the lungs are ventilation, a positive pressure builds up in the lower part of the valve box, and when it exceeds 1.50 mm WG, it presses the control membrane upwards. In consequence, the hinge point of the two levers of the cycling mechanism is drawn above the cycling spring as result of which the floating needle valve and the air-regulating valves are closed and another valve on the junction side is opened. The injector now sucks the air out of the lungs of the patient and blows it into open through a valve situated on the top of the valve-box. As soon as negative pressure of 100mm WG is reached in the lungs, the control membrane is sucked downwards and the cycling mechanism

brought to the position in the inspiration phase. A changing knob is provided to bring the cycling mechanism to expiration position at the beginning of resuscitation.

Points to note:

- During the operation of the Pulmotor, the wearer should always observe that the respiratory system remain constant.
- If, during resuscitation, the action of the heart can no more be observed, heart manage must be carried out during the expiration phase.
- Should serene occurs as hopper with CO poisoning the tongue should be immediately loosened after lifting the face-piece, and if necessary respiration interrupted.
- Pulmotor should not be used except as inheritor in case of poisoning by nitrous fumes.
- It should, however, be emphasized that the first few minutes after breathing has ceased are the most crucial and that manual artificial respiration must be started at once and continue until the Pulmotor becomes available.

Experiment-9

Aim: Evaluation of performance of closed-circuit breathing apparatus using Artificial Lung Machine or Artificial Breathing simulator.

Theory:

Artificial lung machine is also called artificial breathing simulator. It simulates human breathing system. Machine sucks air from self-rescuer where it goes through two different sized piston cylinder arranged, which acts as artificial lung and main lung. Here CO₂ gas is added from as external cylinder to simulate the conditions of human respiratory system in which inhaled air consists of a higher value of carbon dioxide. This CO₂ consisting air is released into the self-rescuer in which the KO₂ granules react with the CO₂ present in the exhaled air and convert it into O₂. During this processes following four parameters are studied.

- (i) Breathing resistance offered by apparatus.
- (ii) O₂ concentration of inhaled air.
- (iii) CO₂ concentration of inhaled air.
- (iv) Temperature of inhaled air.

The test is carried out according to EN 401 which is European Norms related to “Respiratory Protective devices-self-contained closed circuit breathing apparatus for escape-Requirement, testing, marking”, confirming IS 15803.



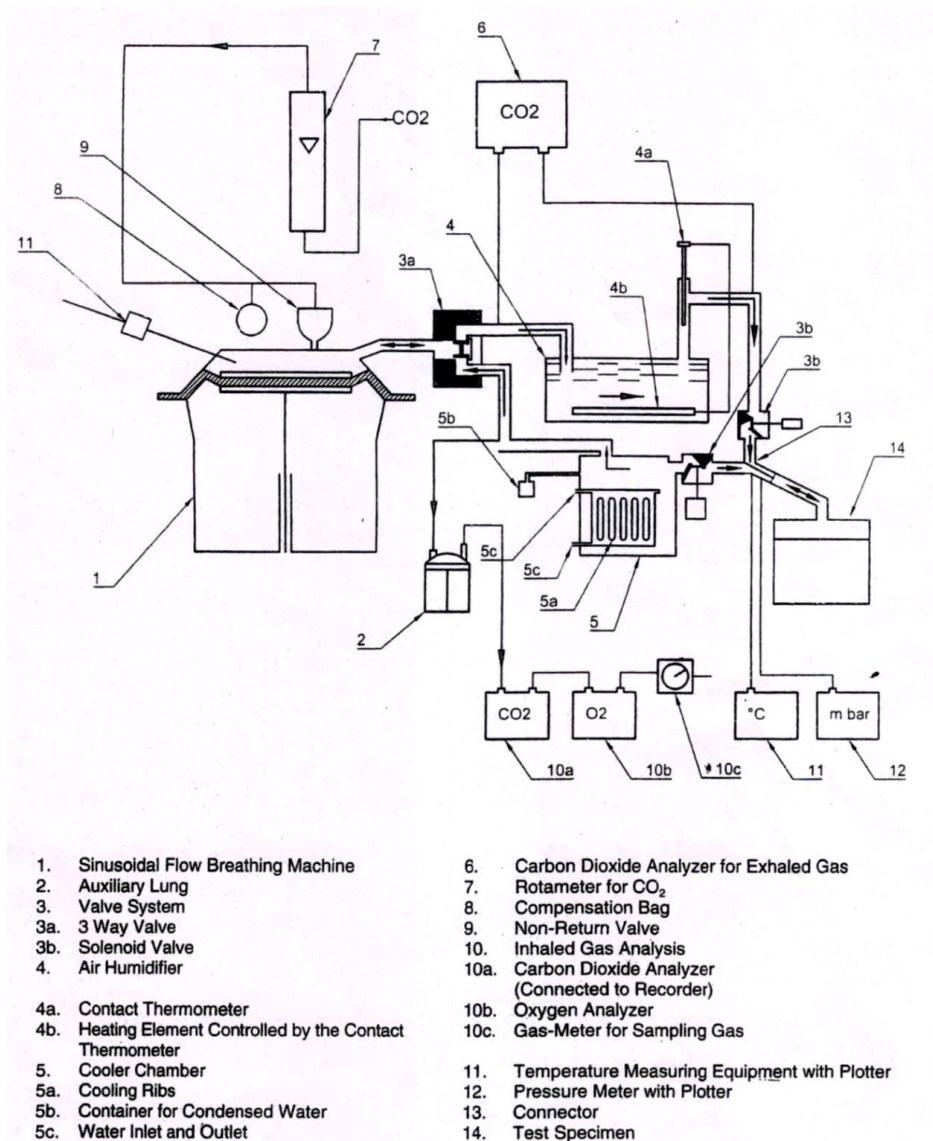


Figure: Photograph and schematic diagram of Breathing Simulator

Test procedure:

Test is carried out in 3 batches, each batch consists of 5 samples. Few special features of test are as follows:-

- (i) Mass of complete apparatus including carrying case or container shall not exceed 5kg.
- (ii) Leak Thickness – The complete design checked by immersing in water.
- (iii) The opening pressure of relief valve shall not be less than 1 milli bar at any condition.
- (iv) Breathing bag capacity or effective volume should be at least 6 ltr.

- (v) The rated duration of apparatus shall be at least 3 times longer than the rated capacity when tested with 10 l/min.

Important Notes:-

1. O₂ content in inhaled air shall not be below 21% by volume and a short term deviation to a level at not less than 17% by volume and for a period of not exceeding 2 min. at the beginning of the test is permissible.
2. CO₂ content throughout the rated duration of apparatus in inhaled air shall not exceed an average value of 1.5% by volume and shall at no time exceed 3% by volume.
3. In the apparatus with a rated duration of 15min., the temp. of inhaled gas shall not exceed 55⁰C when tested at 35 lt/min. during the rated duration.
4. Breathing resistance:-
 - (i) @35 lt/min. – For an apparatus of rated duration of up to and including 30min. the sum of the inhalation and exhalation resistance shall not exceed 16 milli bar. Max. individual breathing resistance shall not exceed 10 milli bar.
 - (ii) @70 lt/min. – For apparatus of all rated duration the breathing resistance for inhalation as well as exhalation shall not exceed 20 milli bar.